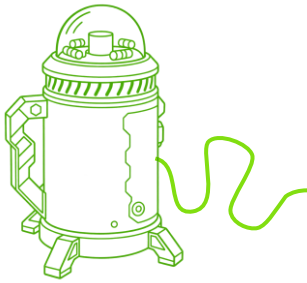
Allsky GO



Assembly Manual (Gen2)

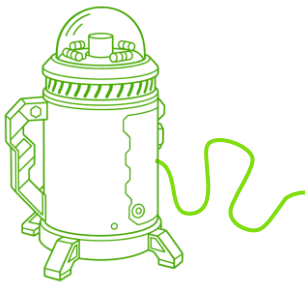
October 2026 v1.0





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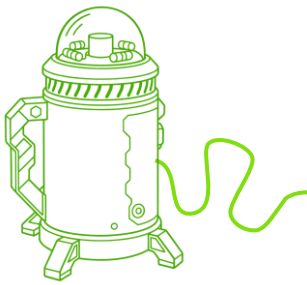




3D Printed Design & Layout

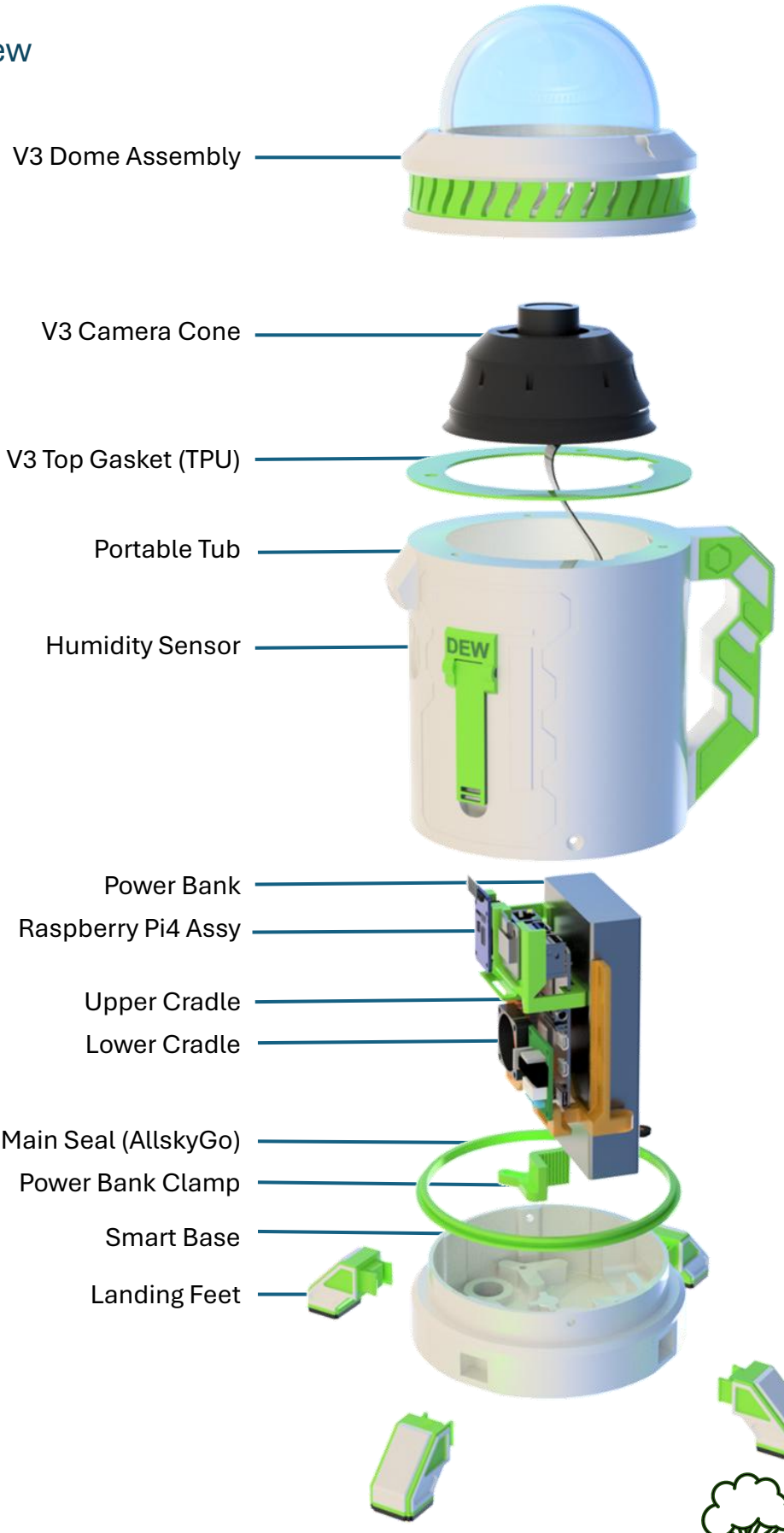








Exploded View



Tub Assembly



V3 Dome Assembly

PiHQ



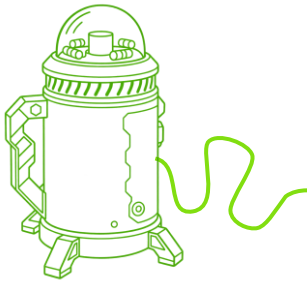
Secure with 3x M3x40-45 Screws
& 4 x M2 Screws for the Camera

=
=
=
=
≠

ZWO



Secure with 3 x M3x40-45 Screws
& 2 x M3 Screws for the Camera





V3 Dome Assembly

No support required for the Standard Dome Assembly. The dome consists of;

- Dome Ring
- Ventilation Housing
- Ventilation Ring
- Camera Cone (ZWO/PIHQ Specific)
- Gasket Seal

Acrylic 3.5” Dome shown from JMX Optical to be purchased separately



Figure 1:Dome Assembly – Pre-Release Pi HQ Version

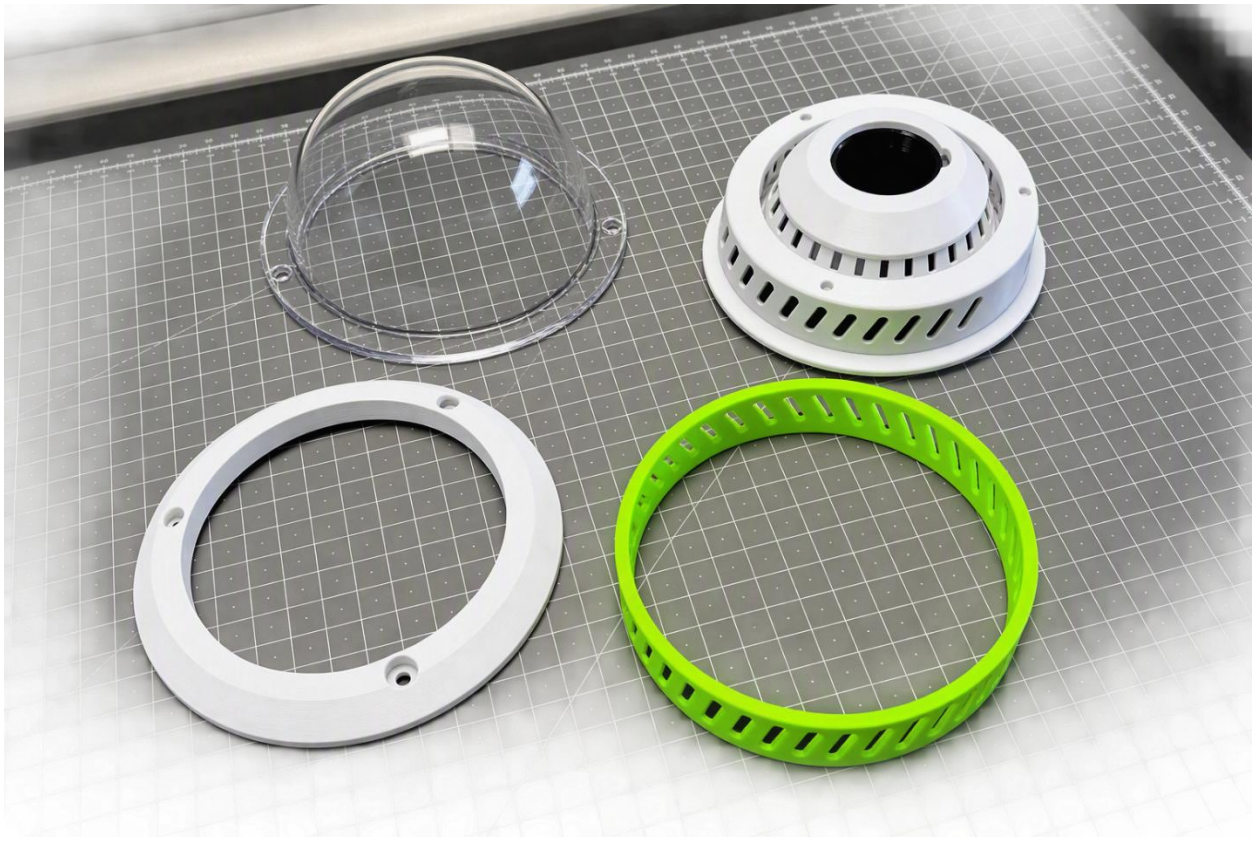


Figure 2:Dark & Mat PETG or ASA/ABS color recommended for the camera Cone to reduce reflections in bright light

Camera Assembly (PiHQ)

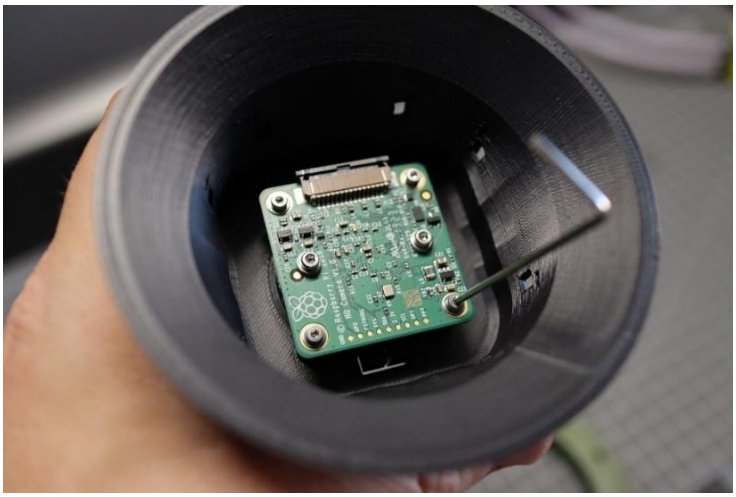


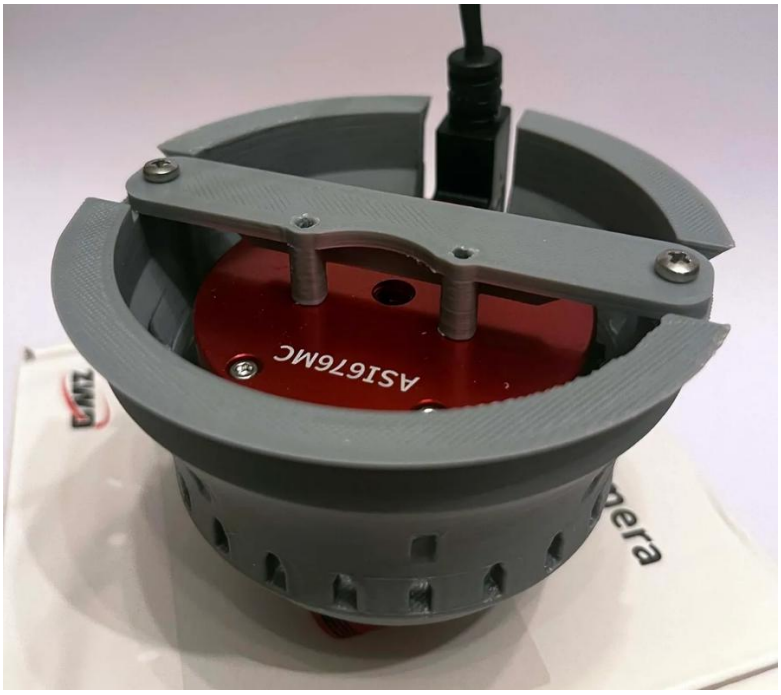
Figure 3: Camera Assembly (Raspberry PI HQ) – 4 x M2 Machine Scews





Figure 4: V3 New Ribbon Guide! Facilitates extraction of Ribbon without having to dis-assemble the camera

Camera Assembly (ZWO)



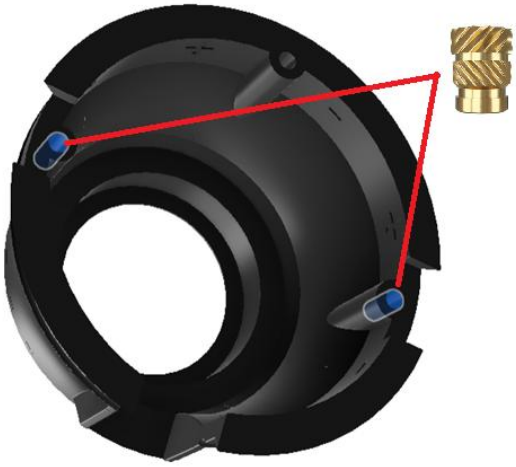


Figure 5: Camera Assembly (ZWO) with Retainer and 90deg. USB Connection

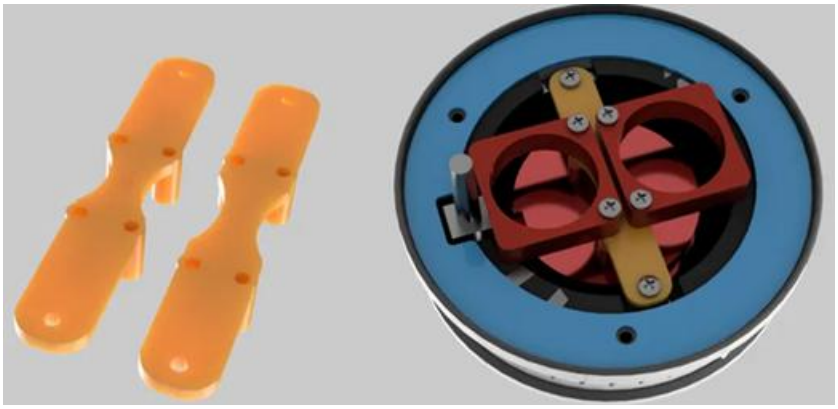


Figure 6: Retaining Brackets double as Cooling Fan Support (30mm and 40mm)



Figure 7: ZWO Housing with 90˚ USB3 TypeB Connector

Dome Heater Assembly (Optional)

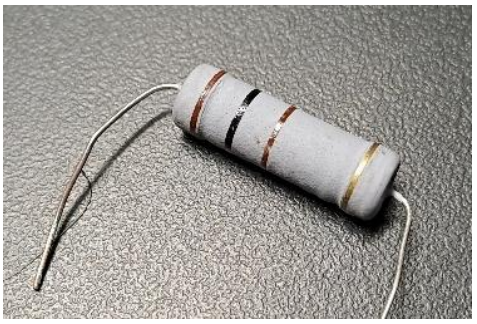


Figure 8: 100ohm Metal Oxide Film Resistors rated for 5W

Resistor Set-up

1W Heating Solution always on without heating control:

- 4 x 100ohm Metal Oxide Film Resistors, Rated for 5W,
- Max Power 1W connected in Parallel (5V, 200ma)
- USB Power Cable Cut & soldered to GPIO Jumpers

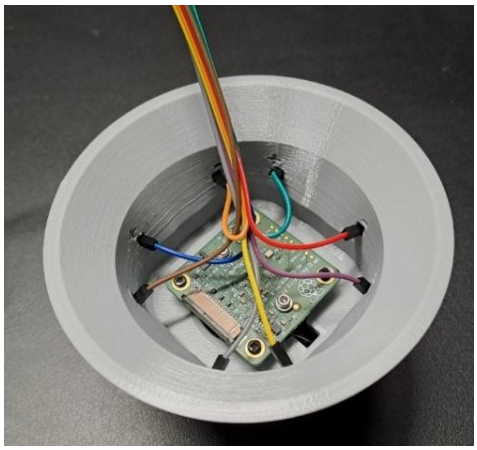


Figure 9: Harness around camera module



Figure 10: Resistors Lock into Female Jumper Wires

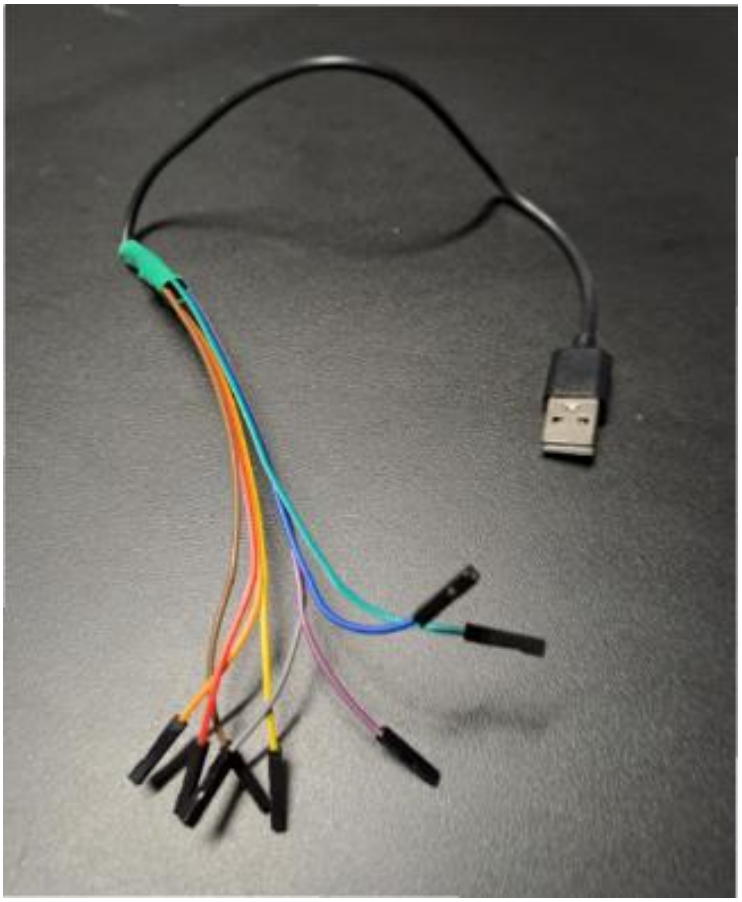


Figure 11: USB Source Split into 4 positive + 4 negative leads

Assessing the Resistors

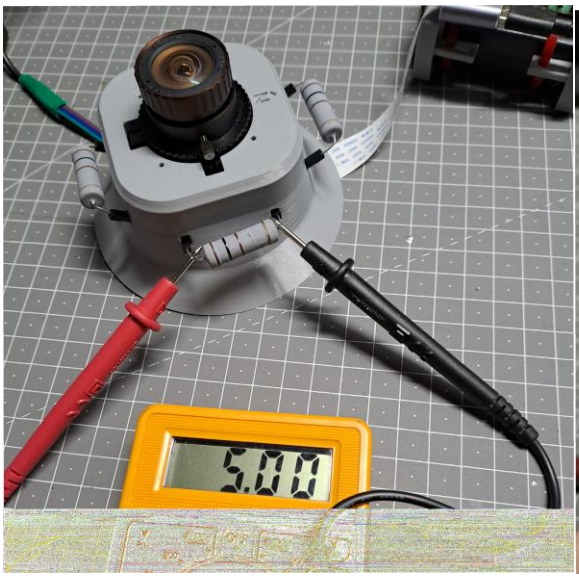




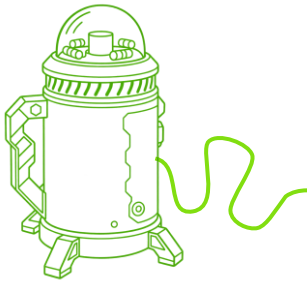
Figure 12: To check connection in parallel; each resistor should receive 5volts –

Figure 13: Measured Temperature range around +10° ambient (37°C max recorded)

Raspberry Pi USB Power Output Comparison

The selected Power source may limit the USB Power Out capability of your Raspberry Pi – and as such the power to your resistors or function of USB peripherals. Resistors should draw 200ma for 1W total power at 5V.

Model	***Max USB Power Output (Combined)***	***Per-Port Limit***	***Notes***
Raspberry Pi 3	*1.2A total across four ports*	*No per-port limit*	*Uses AP2553 chip to limit total USB current. Requires adequate PSU.*
Raspberry Pi 4	*1.2A total across four ports*	*No per-port limit*	*USB 3.0 ports available. Needs 3A PSU to fully support USB draw.*
Raspberry Pi 5	*1.6A total across four ports*	*600ma per-port limit*	*Requires 5V/5A USB-C PD PSU or config override to unlock full output.*





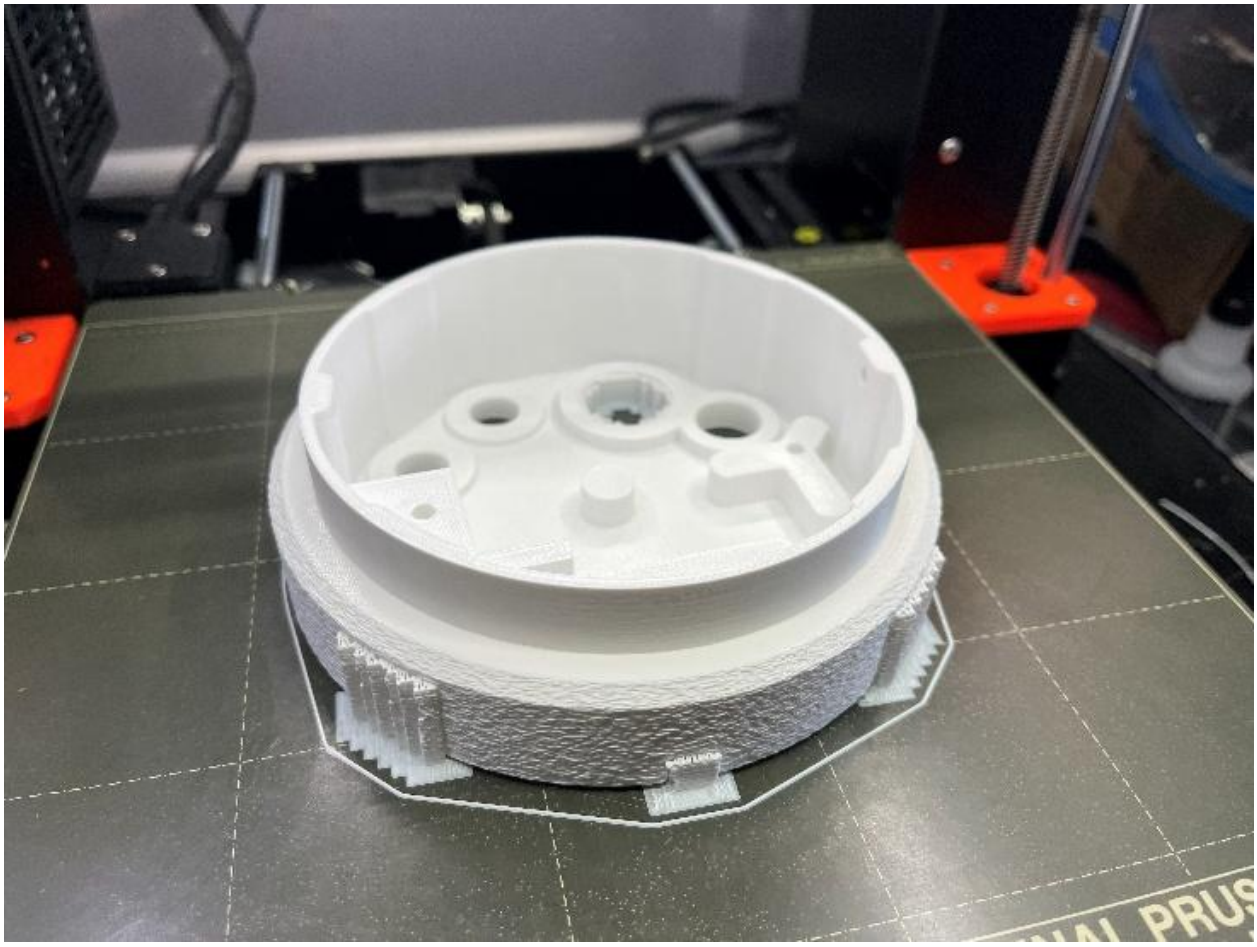
Printing the Main Housing Components

Printed with 0.4 Nozzle

Print on **smooth Print bed** to optimize sealing surfaces.

If printing in ABS or ASA – please be cautious of eventual **warping** on large parts that can impact the flatness of sealing edges. Additional Tabs to increase bed adhesion may be required.

Printing with selective **Fuzzy skin** on the external surfaces can reduce the visibility of print lines, or variable layer height.



Install the M3 and 1/4-20 Inserts with soldering iron

M3 Insert

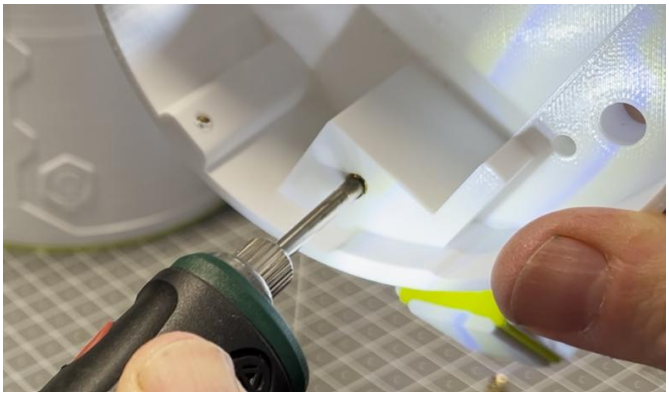
Qty 2 in Side; Tub Fixation

Qty 2 in Top; Cradle & Clamp Fixation

Qty 4 in Base (optional for studs)

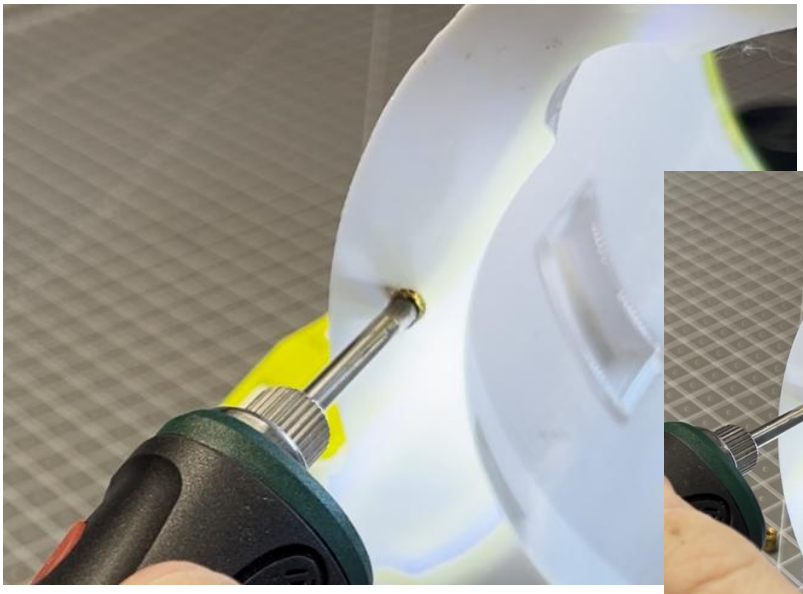
¼-20 Tripod Insert

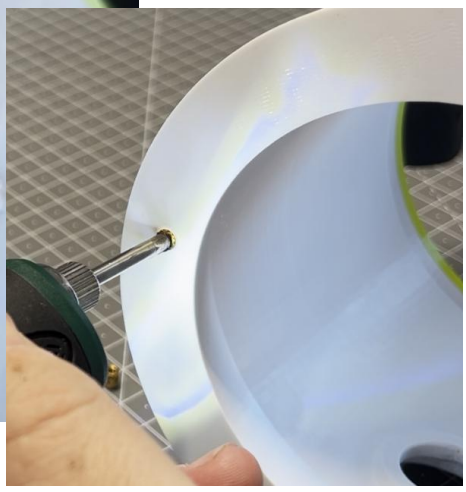
Qty 1 in Base





Qty3 Inserts on top of main Tub;

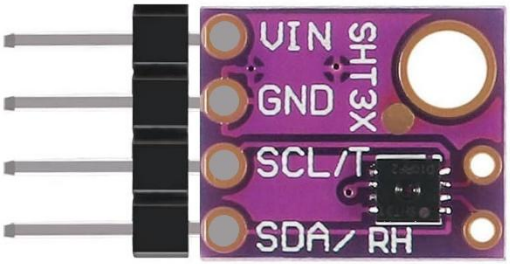


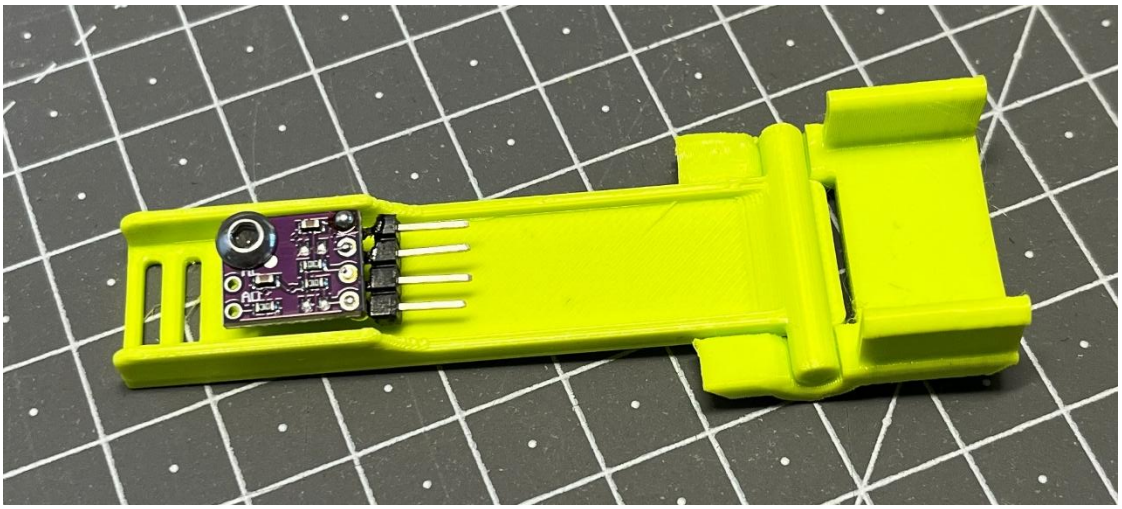


Humidity Sensor Probe

Solder pins flat against the four terminals

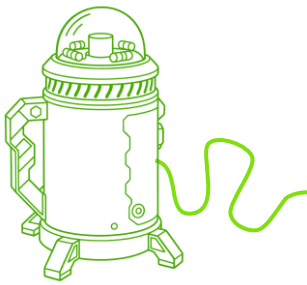
Qty1 M3x4mm Screw for fixation to probe.





Pass Qty 4 GPIO Jumpers through tub opening (Female-Female)

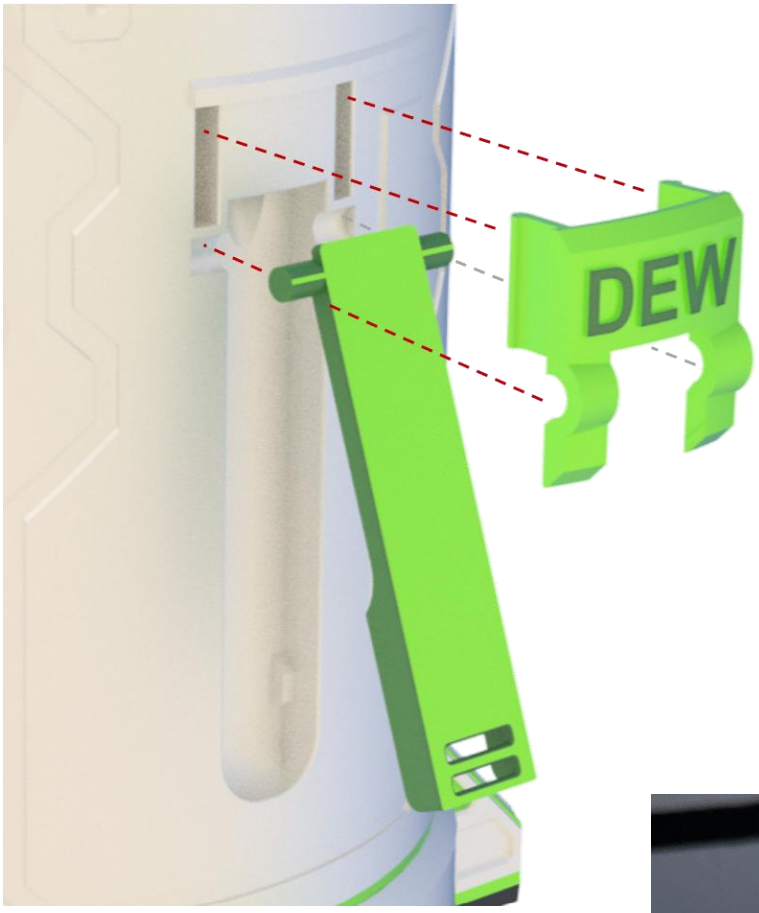


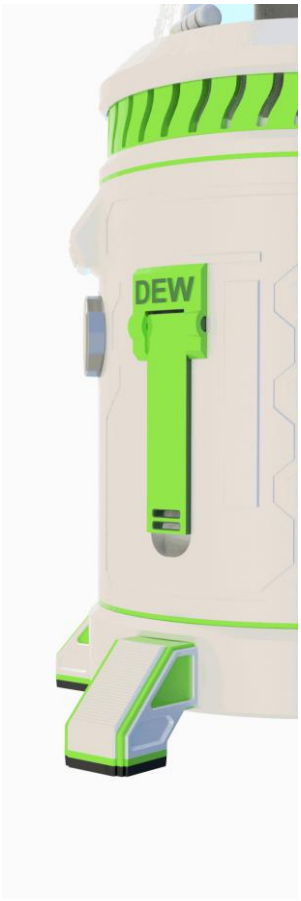


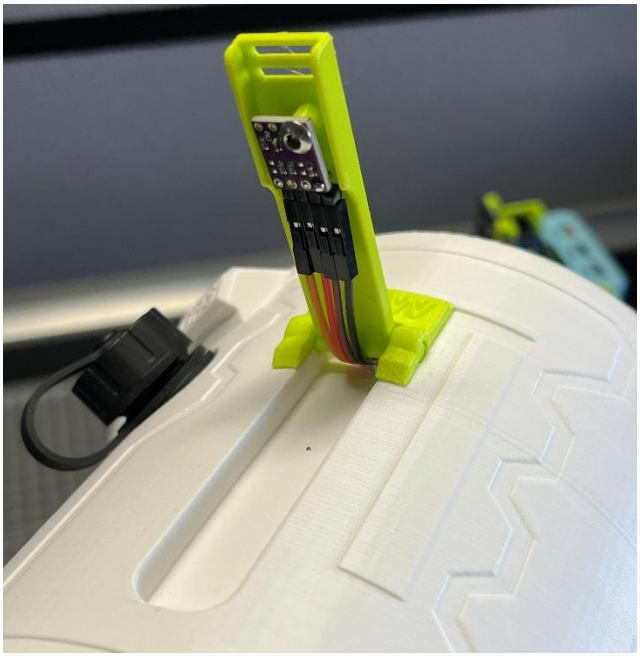


Humidity Probe push-fits into side of the tub

No screws or glue required



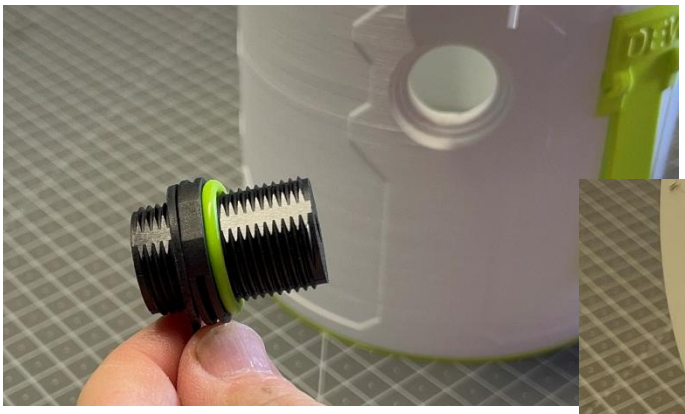


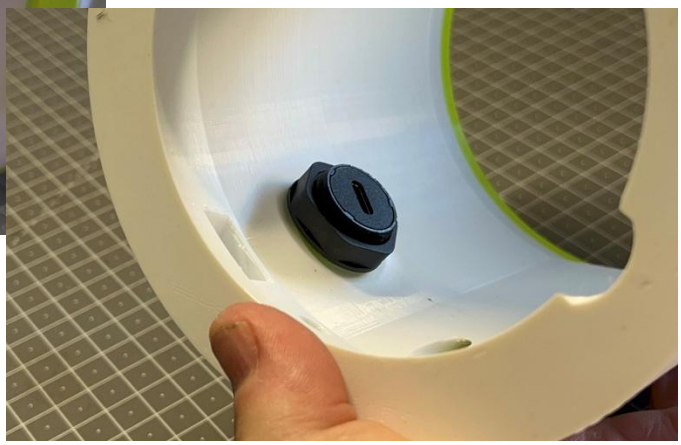


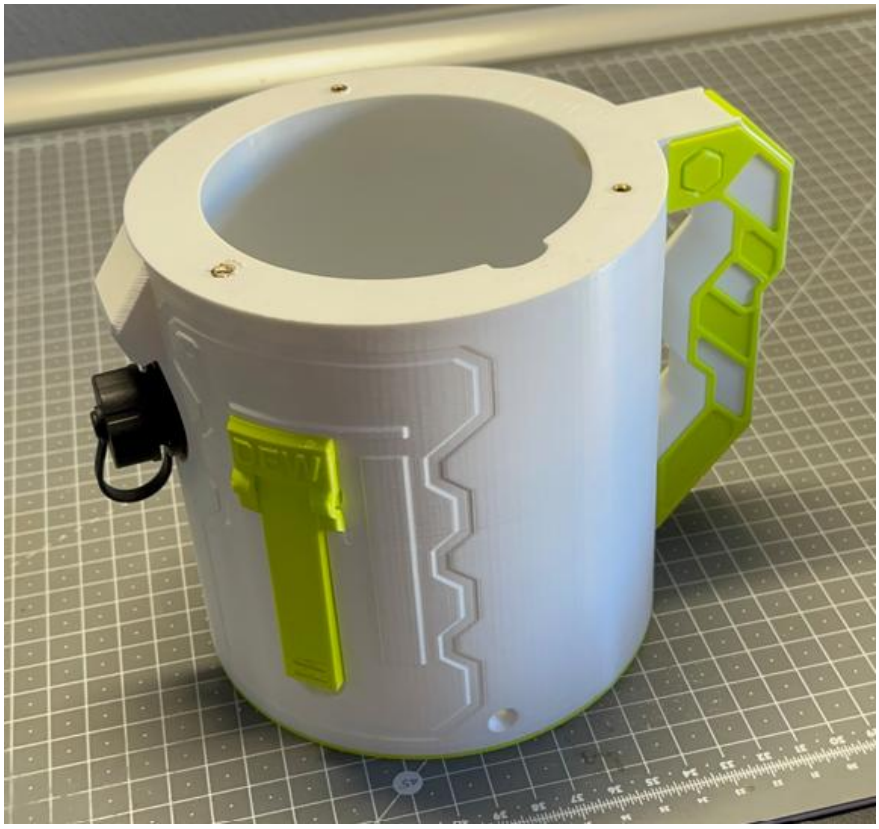
Tub Finalization

Prepare the USBC Pass through – and insert into the port.

Ensure the O-rings are positioned correctly to avoid water ingress, and secure with the end cap.

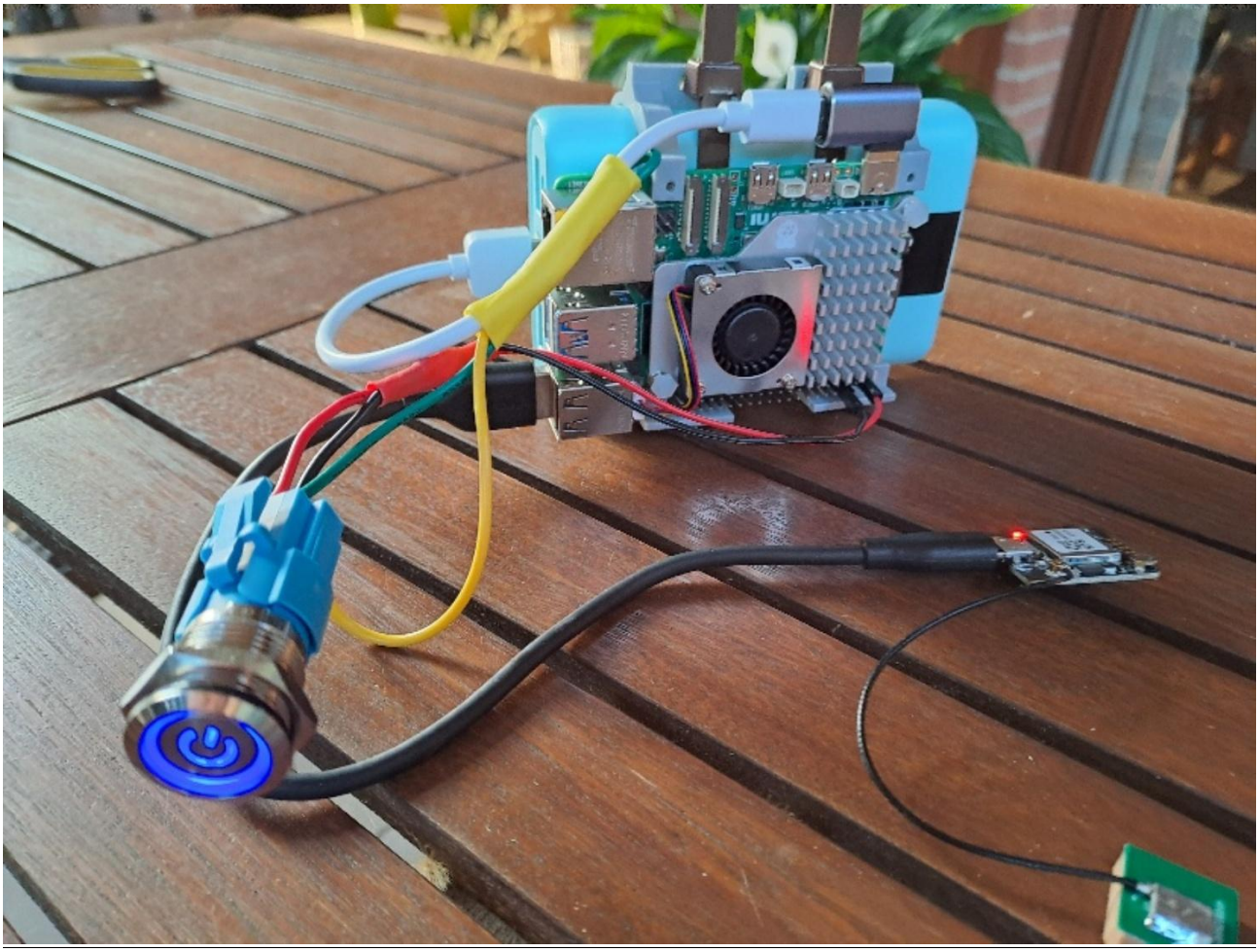






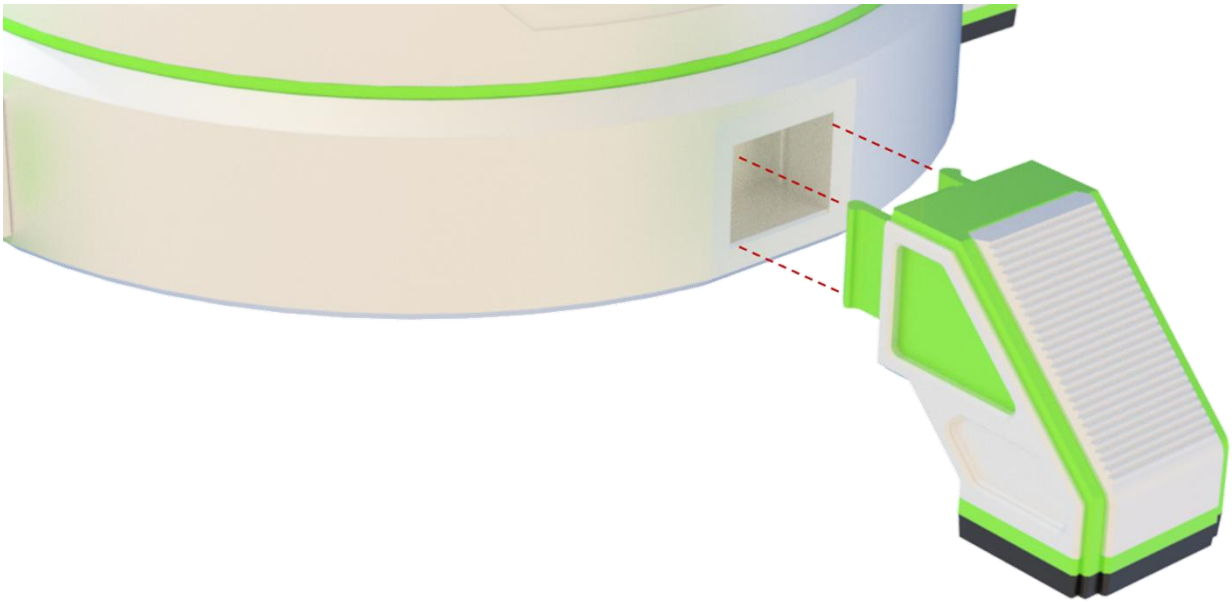
Electronics Check(!)

Before the final assembly, check the harnesses and electronics are wired and functioning properly, GPIO Pins are correctly arranged and Allsky software and scripts running reliably.

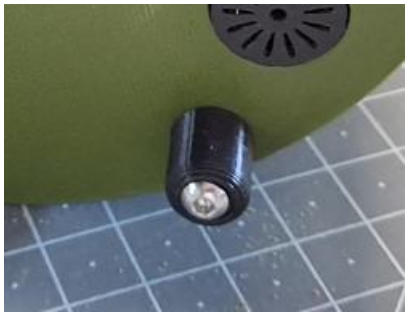


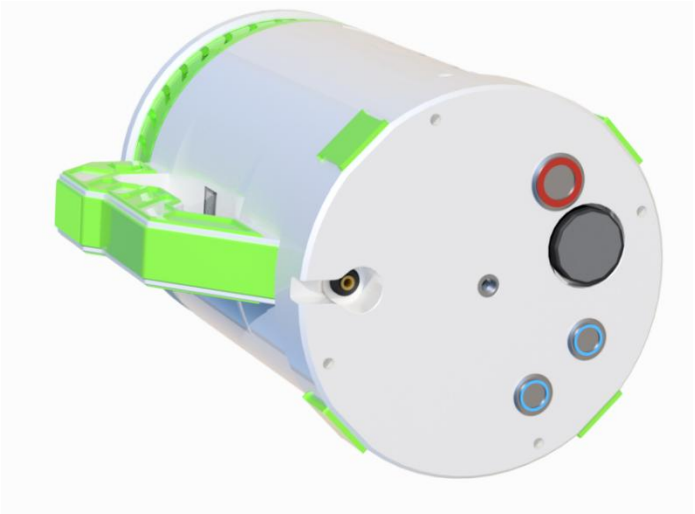
Base Installation

Push Fit Landing Legs



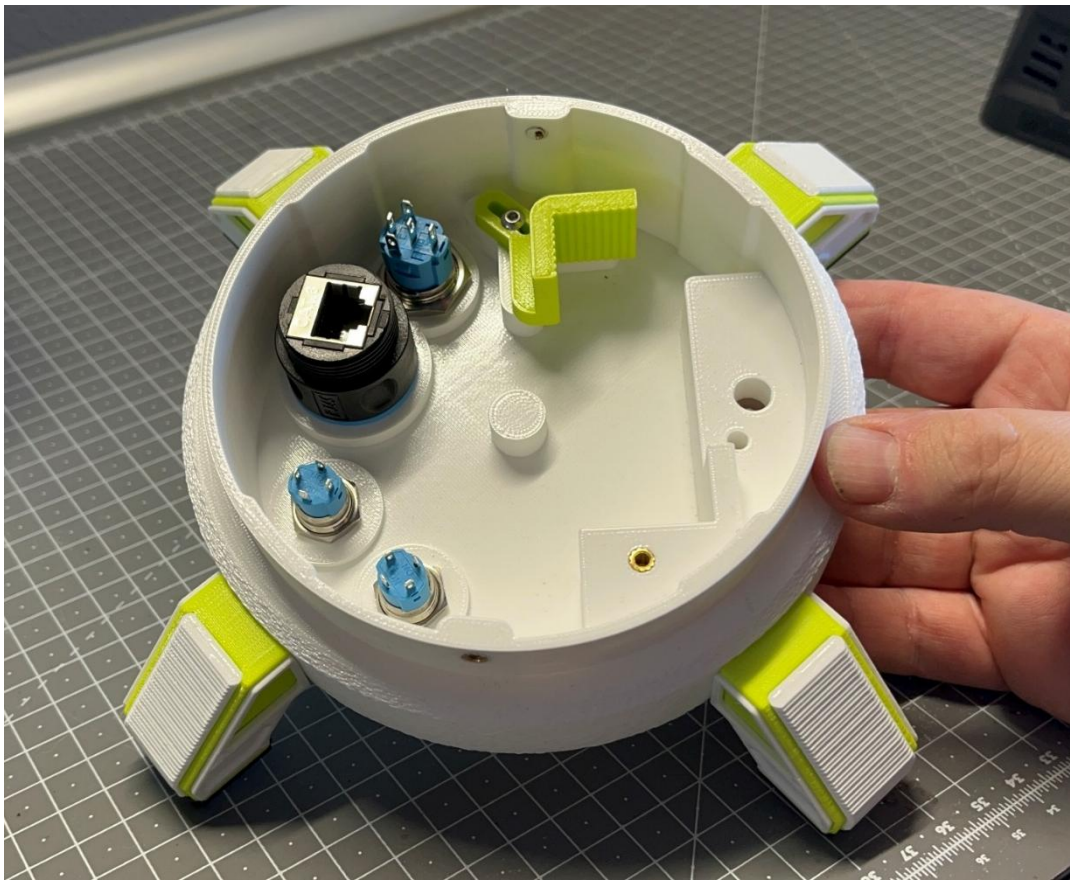
4 Studs using M3 screws and inserts remain an alternative base arrangement.
Can be used in combination with the plugs to fill the feet sockets for a clean side profile.





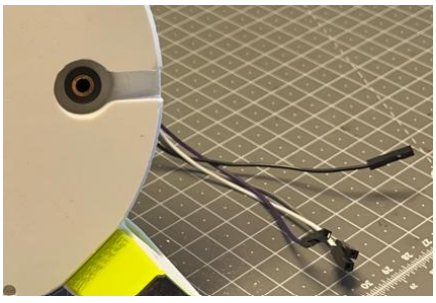
Install the main switches and ports.

The M12 and M16 switches come provided with their own seals to protect against rain or moisture. The RJ45 port threads directly into the base.



The optional female jack plug cut-out can be drilled through at this point. This 4-pole connection can be used for additional sensors.





Battery Clamp

Install the battery clamp with M3 screws and check the fit of the Powerbank.

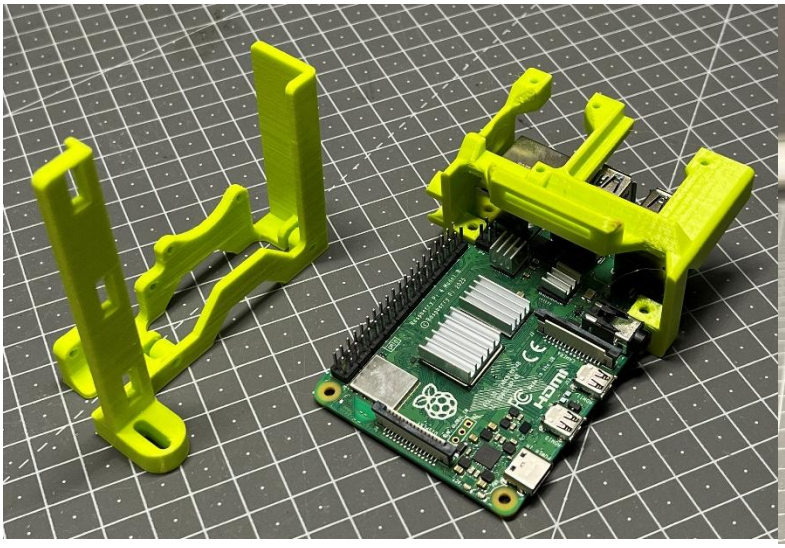


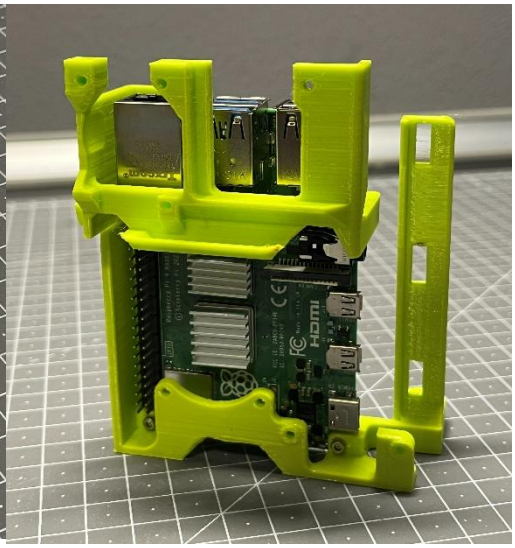
Cradle Assembly



Insert the Pi4B into both Upper and Lower Cradle

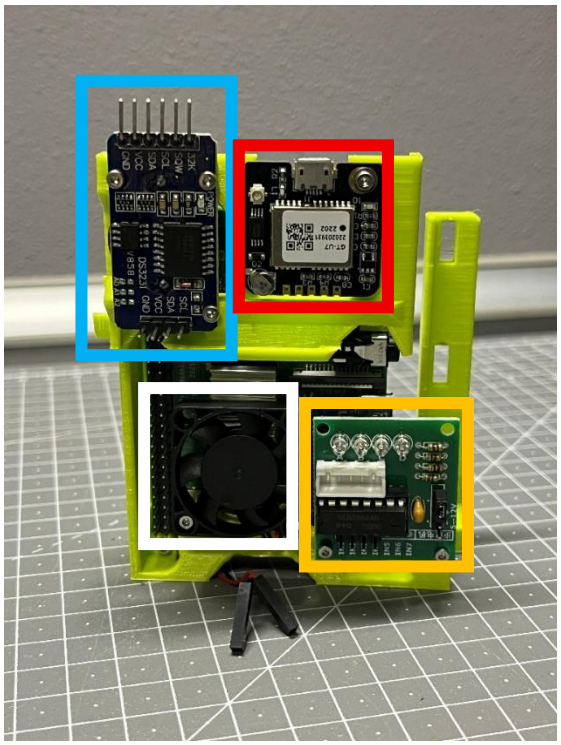
Fix with Qty 4 M2x7mm screws





Hardware

Install the Hardware to the cradle Assembly:



FAN Qty 2 M2 x 7mm

GPS Antenna Qty1 M2 x 5mm (with washer)

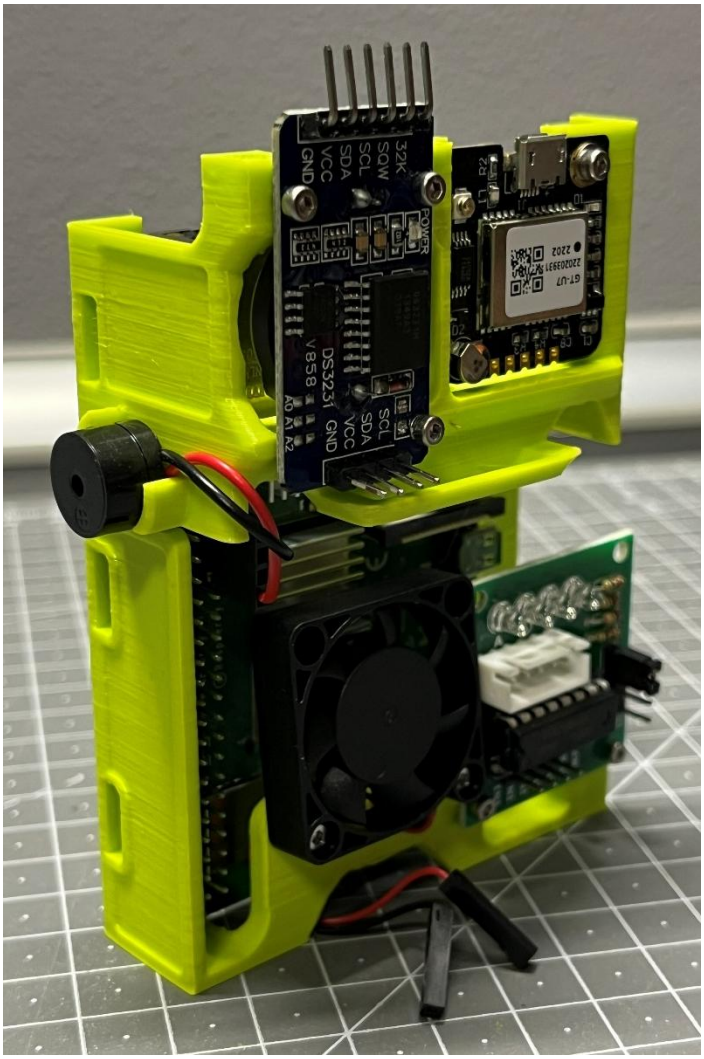
DS3231 RTC Module Qty2 M2 x 5mm

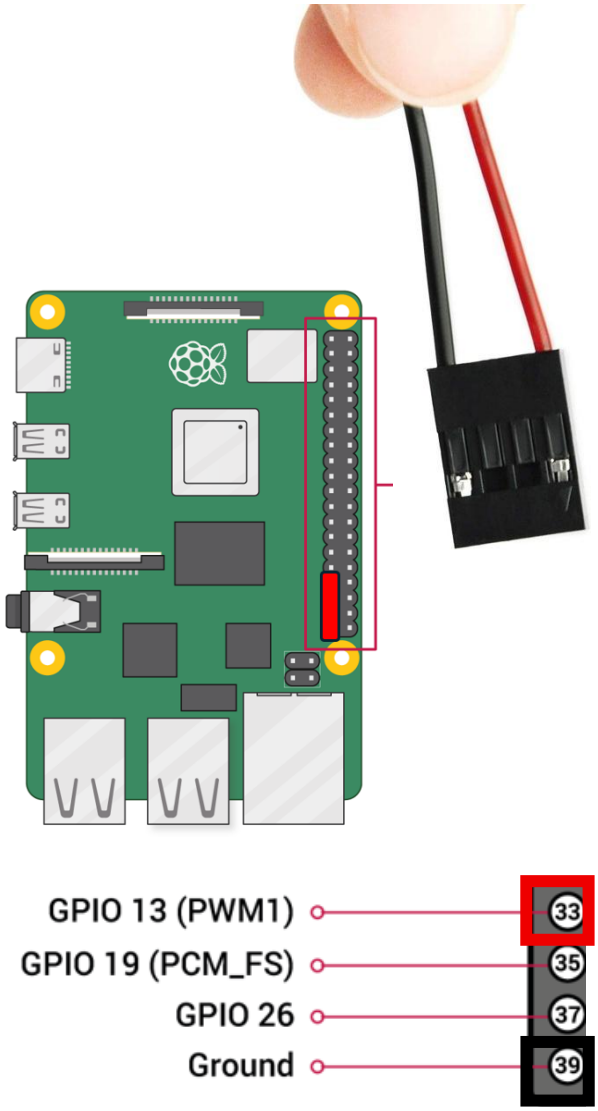
ULN2003 Qty 2 M2 x 5mm

(Optional, for External Motor Control)

Piezo Buzzer

Tuck the 12mm Piezo Buzzer within the side clamp.
The plug can be installed to the last 4 pins (GPIO13 + **GROUND**)





Cradle Installation

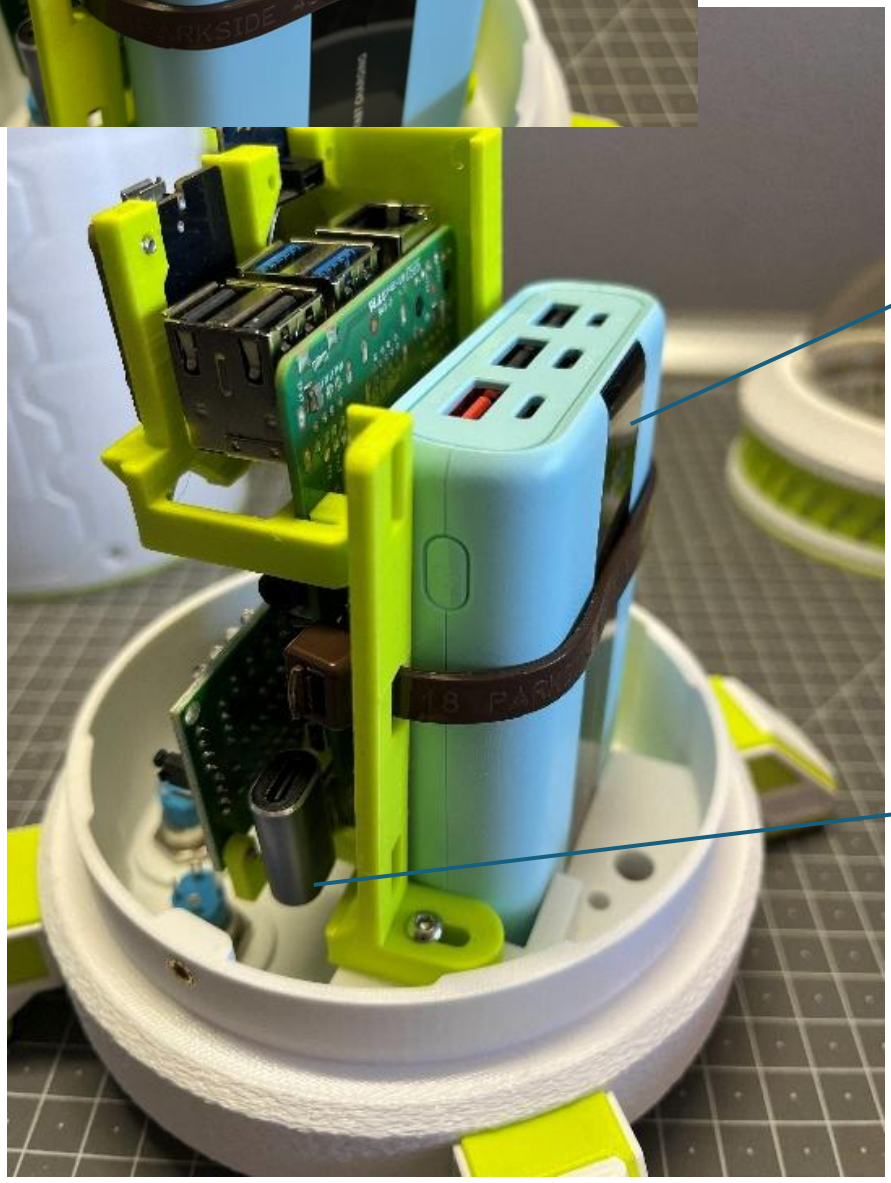
Installed the cradle with a M3 screw – the cradle should be flush against the powerbank.



The Powerbank can be fixed with **cable ties** to secure against the cradle assembly. Install the PIHQ **Ribbon Cable** here, passing behind the GPS module.











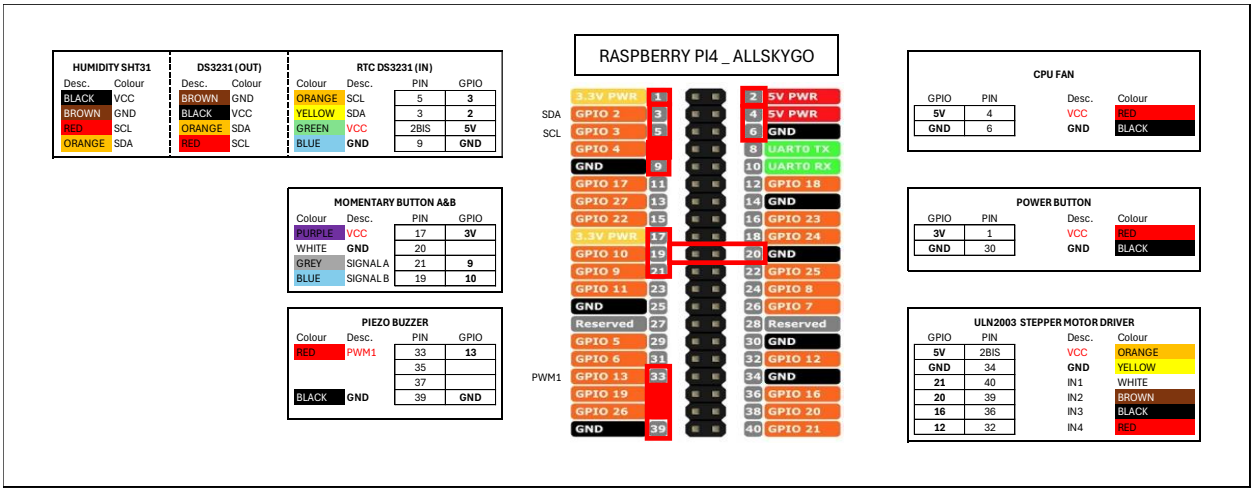
Wire Harnesses and Routing

See wiring schematics for detailed proposal.

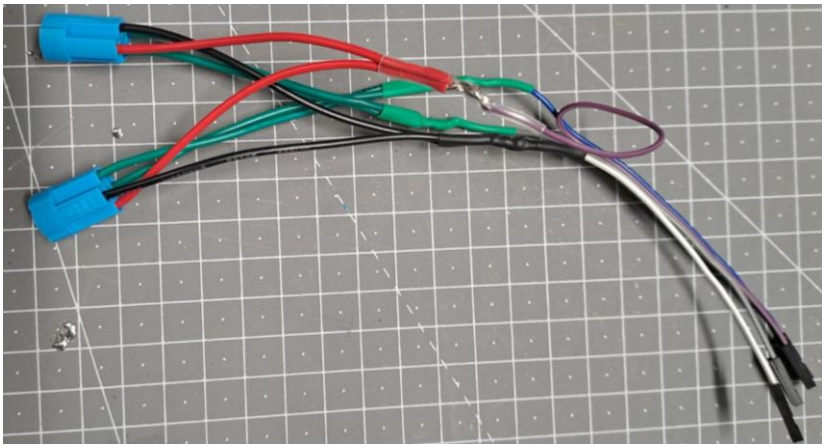
Harnesses are constructed from standard jumper wires and short 15cm USB Cables.

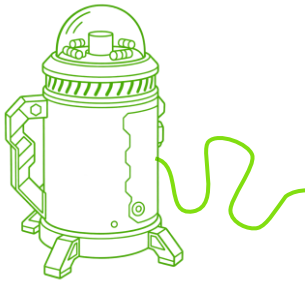
Avoid to block the airflow around the fan.

Attached Excel to support mapping of the harness according to colour of wires available.

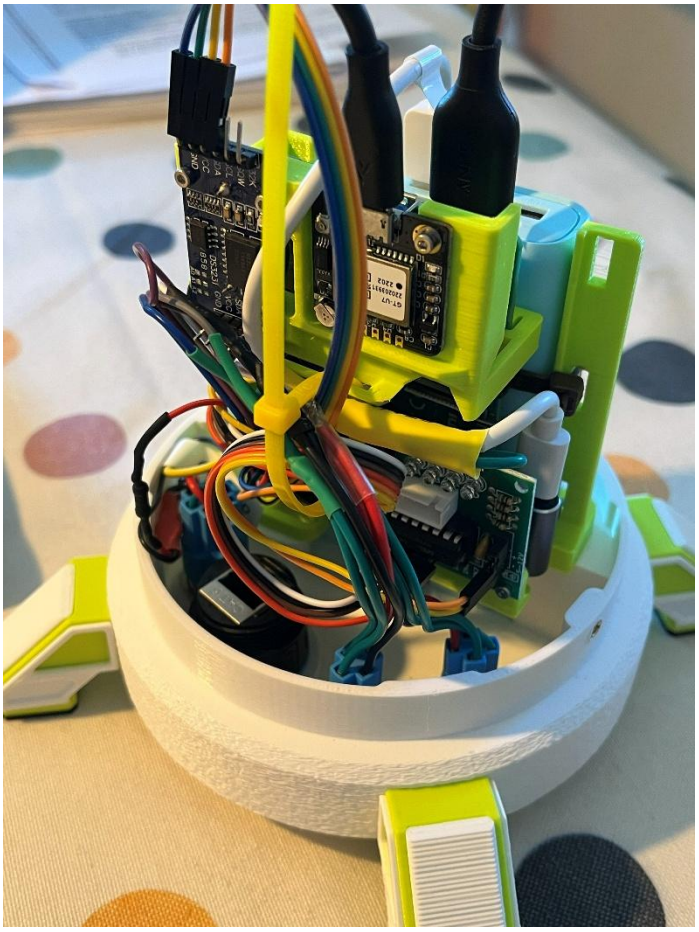


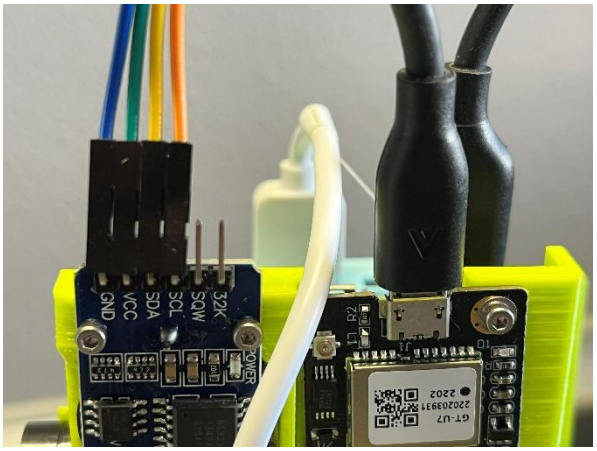
Example Button Harness

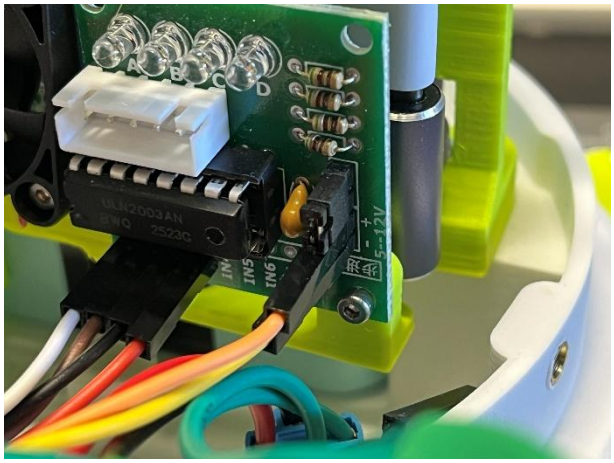






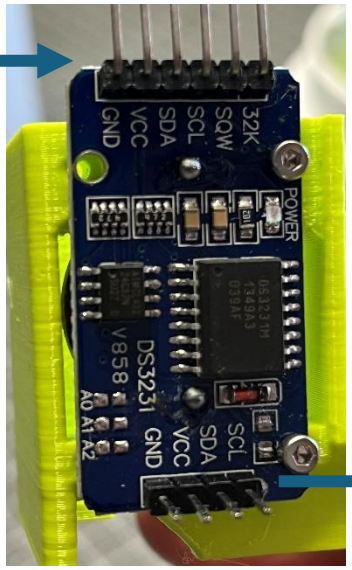






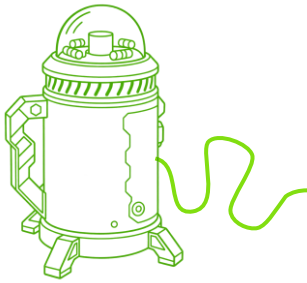
I2C Daisy Connection RTC > SHT31

Input from PI





Output to SHT31 Humidity Probe

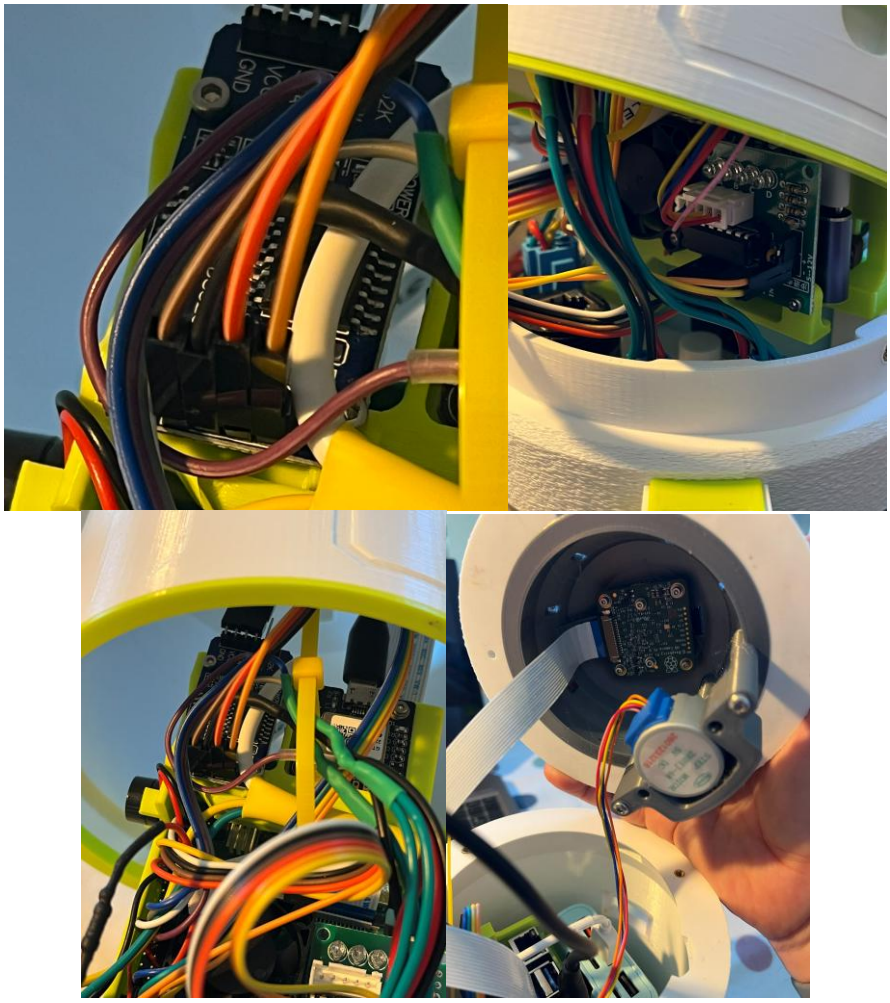




Closing the Tub

Depending on your set-up this may be more or less tricky. Ensure pins are correctly secured and slide over the main tub assembly.

- Connect the camera ribbon cable
- Connect the SHT31 Sensor to DS3231 board
- Connect the USB extension to Powerbank power Input
- Connect GPS Antenna and insert into available Slot
- Connect the USB heating (Optional)
- Connect the stepper motor (Optional with Focus Pack)

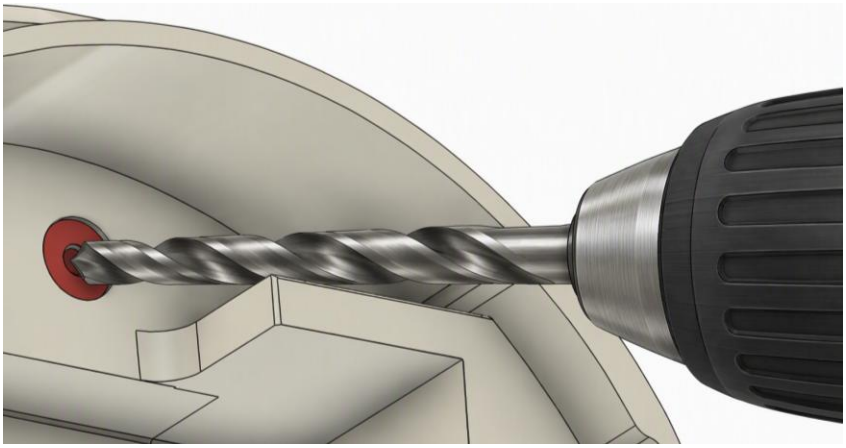


External Sensors

Within the base – there is the opportunity to install a 4 Pole Jack plug for external sensor connections. Based on 4pole- jack plug insert available on Amazon: https://amzn.eu/d/018eTmEq



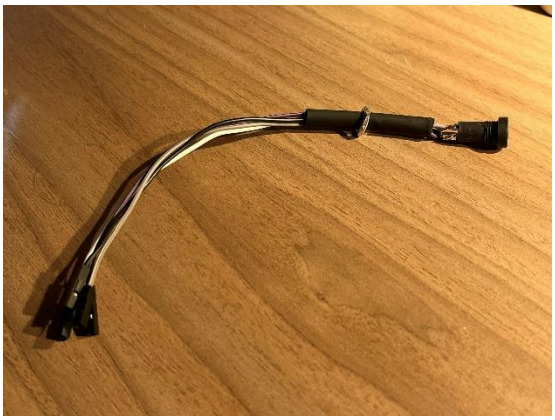
Requires 8mm cut-out in slicer, or 8mm drill bit post printing.



The port is retained by the nut provided.

Standard jumper cables can be soldered to extend to GPIO Board.

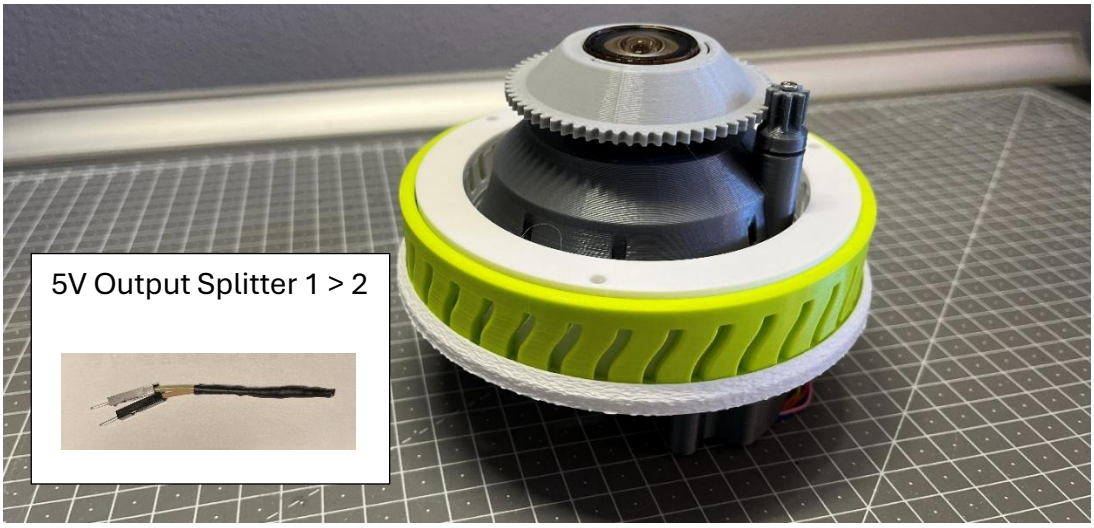




V3 Dome Focusing Upgrade

(Requires Optional Upgrade Pack, Stepper Motor + Driver)

May require additional split of 5V rail



V3 USB Dew Heating

After Assembly, Complete installation of Resistance within Dome Assembly.






DEW